

Tiancheng LIU

Ph.D. Candidate at The Hong Kong University of Science and Technology (Guangzhou)

Visiting Researcher at University of Amsterdam

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ABOUT ME

Tiancheng Liu is a computational media researcher and artist working across Creative AI, multimodal interaction, and culturally grounded computing. His work investigates how machines can understand, sense, and participate in human creative practices, with a particular focus on embodied cultural forms such as Chinese calligraphy. His Ph.D. research is organized around three interrelated layers of creative practice. The first layer asks how machines can analyze and interpret the aesthetic structure of finished artifacts, including layout, form, balance, and stylistic preference. The second layer asks how machines can sense the often invisible processes behind creation, such as gesture, pressure, rhythm, and other implicit micro-actions. The third layer asks how AI systems can join creative practice as collaborators, supporting co-creation, performance, and expressive interaction. Building upon this framework, he seeks to explore broader possibilities for computing approaches as a medium for interpretation, expression, and collaboration, expanding how humans and machines can sense, make, and imagine together in the cultural and artistic context.

EDUCATION

University of Amsterdam , Amsterdam, Netherlands	04.2026 – Present
Visiting Researcher in MultiX, Informatics Institute	
HKUST (GZ) , Guangzhou / Hong Kong SAR, China	09.2022 – Present
Ph.D. Candidate in Computational Media and Arts, Information Hub	
Waseda University , Tokyo, Japan	09.2018 – 10.2020
M.Eng in Information	
South China University of Technology , Guangzhou, China	09.2014 – 07.2018
B.Eng in Network Engineering	
South China University of Technology , Guangzhou, China	02.2015 – 07.2018
B.Econ in Finance	
Zhixin High School Guangzhou, China	09.2011 – 06.2014

PUBLICATIONS

Accepted

- [1] Tiancheng LIU, Nanne Van NOORD, Chen LIANG, and Kang ZHANG. “Full-Page Chinese Calligraphy Character Detection: A Dataset and Benchmark with Integrity-Preserving Tiling and SDAC-Det”. In: *VISART VIII: Vision for Art and Culture Workshop, European Conference on Computer Vision (ECCV)*. Accepted; DOI TBD. Malmö, Sweden, 2026.
- [2] Tiancheng LIU, Shumeng ZHANG, Hanyu QU, and Hanqin ZHOU. “NarraStone: Reimagining the Stone Stele as an AI-Era Apparatus of Synthetic Memory”. In: *Proceedings of the SIGGRAPH Asia 2026 Art Papers*. SA Art Papers '26. Accepted; DOI TBD. Kuala Lumpur, Malaysia: Association for Computing Machinery, 2026.

Published

- [1] **Tiancheng Liu**, Shumeng Zhang, and Nicolò Merendino. “BruSHŪ: Cross-Modal Translation of Implicit Micro-Actions in Chinese Calligraphy”. In: *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 9.3 (2026). To be presented at SIGGRAPH 2026, Los Angeles, CA, 19–23 July 2026, pp. 1–13. [\[Online\]](#).
- [2] Piera Riccio, Selina Khan, Ludovica Schaerf, Shuai Wang, **Tiancheng Liu**, Athanasios Efthymiou, Noa Garcia, and Nanne van Noord. “Understanding Art & Culture”. In: *Proceedings of the 2026 International Conference on Multimedia Retrieval*. ICMR ’26. Association for Computing Machinery, 2026, pp. 2901–2903. [\[Online\]](#).
- [3] Shumeng Zhang, Shiqi Lin, **Tiancheng Liu**, Mingming Fan, and Raul Masu. “To Perform/To Live: Decolonizing Digital Music Instruments and Feminism with Human-AI Co-Created Embodied Experience of Daily Objects”. In: *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 9.3 (2026). To be presented at SIGGRAPH 2026, Los Angeles, CA, 19–23 July 2026, pp. 1–12. [\[Online\]](#).
- [4] **Tiancheng Liu**, Siyu Yan, Shumeng Zhang, Chen Liang, and Kang Zhang. “Computational Interpretation of Chinese Calligraphy Layout via Graph-Based Modeling and Language Models”. In: *Proceedings of the POLYU COMP - HKUST(GZ) INFH Research Student Conference*. Poster presentation. Hong Kong, China, 2025.
- [5] **Tiancheng Liu**, Jiayi Ye, Shumeng Zhang, Kang Zhang, and Chen Liang. “Quantifying Structural Aesthetic Features and Personality Trait Preferences in Kai Shu Calligraphy”. In: *Proceedings of the 33rd ACM International Conference on Multimedia*. MM ’25. Dublin, Ireland: Association for Computing Machinery, 2025, pp. 6730–6739. [\[Online\]](#).
- [6] Yinzong Wei, Yiming Fu, Siqi Wang, Shuwei Gao, **Tiancheng Liu**, Zeyu Wang, and Xin Tong. “A Deep-Learning Model for Edition Identification of Premodern Chinese Rare Books”. In: *Digital Humanities* 03 (2025). [\[Online\]](#), pp. 1–10.
- [7] Shumeng Zhang, **Tiancheng Liu**, and Mingming Fan. “FinHERtip: Embodied Identity and Human-AI Co-Creation in Accessible Musical Performance”. In: *Proceedings of the SIGGRAPH Asia 2025 Art Papers*. SA Art Papers ’25. Association for Computing Machinery, 2025. [\[Online\]](#).
- [8] **Tiancheng Liu**, Ling Li, and Manling Yang. “Chinese Seal Carving Aesthetic Evaluation”. In: *Proceedings of the 17th International Symposium on Visual Information Communication and Interaction*. VINCI ’24. Association for Computing Machinery, 2024. [\[Online\]](#).
- [9] **Tiancheng Liu**, Anqi Wang, Xinda Chen, Jing Yan, Yin Li, Pan Hui, and Kang Zhang. “PoEmotion: Can AI Utilize Chinese Calligraphy to Express Emotion from Poems?” In: *Proceedings of the 29th International Symposium on Electronic Arts: ISEA2024 Everywhen Proceedings. Volume 1: Academic Papers*. Ed. by Gavin Sade, Andrew Brown, Leah Barclay, Jen Seevinck, Anastasia Tyurina, and Rewa Wright. Brisbane, Qld: Queensland University of Technology and ISEA International, 2024, pp. 476–483. [\[Online\]](#).
- [10] Aven Le Zhou, Jiayi Ye, **Tiancheng Liu**, and Kang Zhang. “Archiving Body Movements: Collective Generation of Chinese Calligraphy”. In: *Proceedings of the 29th International Symposium on Electronic Arts: ISEA2024 Everywhen Proceedings. Volume 1: Academic Papers*. Ed. by Gavin Sade, Andrew Brown, Leah Barclay, Jen Seevinck, Anastasia Tyurina, and Rewa Wright. Brisbane, Qld: Queensland University of Technology and ISEA International, 2024, pp. 862–870. [\[Online\]](#).

- [11] **Tiancheng Liu** and Keiichi Koyanagi. “Blockchain Technology Application in Access Control”. In: *Proceedings of the 6th Waseda ORIS International Seminar for Junior Researchers*. Waseda University. Tokyo, Japan, 2020.
- [12] Helong Li, Tianqi Luo, Liuling Li, and **Tiancheng Liu**. “Measuring Systemic Risk in the Chinese Financial System Based on Asymmetric Exponential Power Distribution”. In: *Recent Developments in Data Science and Business Analytics*. Ed. by Madjid Tavana and Srikanta Patnaik. Cham: Springer International Publishing, 2018, pp. 225–232. [Online].

EXHIBITIONS

Ars Electronica 2026, presenting <i>BruSHŪ</i> , Linz, Austria,	10.2026
SYMBOTIC SIGNALS, presenting <i>BruSHŪ</i> , Guangzhou, China	06.2026
InnoCharity, presenting <i>BruSHŪ</i> , Guangzhou, China	05.2026

GRANTS AND FUNDING SUPPORT

Project Researcher (Proposal Co-author) — National Social Science Fund of China (NSSFC) Major Project

HKUST (GZ) 2024–present

Topic: Application of Frontier Technology in Cultural Heritage Protection and Inheritance

- **Program:** Special Program on Cultural Heritage Protection and Inheritance Research
- **Co-applicants:** PI: Prof. Kang ZHANG
- **Grant:** No. 24VWB020

Recipient — Overseas Research Travel Grant (RTG)

HKUST (GZ) 2024, 2025 & 2026

Topic: Travel to present research at international conferences (outside Greater China)

- **Role:** Recipient
- **Funding:** RTG conference travel support

Recipient — Overseas Study Award

HKUST (GZ) 2026

Topic: Visiting researcher appointment at University of Amsterdam (UvA)

- **Role:** Recipient
- **Funding:** Overseas study support

AWARDS & SCHOLARSHIP

First Prize, Model Creativity Competition, by *Red Bird Maker Space, College of Future Technology, Hong Kong University of Science and Technology (Guangzhou)* 2024

Finalist, Three Minute Thesis (3MT) Competition, by *Hong Kong University of Science and Technology (Guangzhou)*, founded by The University of Queensland 2024

Postgraduate Studentship (PGS), research studentship awarded by *Hong Kong University of Science and Technology (Guangzhou)* 2022 – Present

JASSO Scholarship — *School of Information, Production and System, Waseda University, Tokyo, Japan* 2018 – 2020

Second-Class Academic Scholarship — *School of Computer Science and Engineering, South China University of Technology, Guangzhou, China* 2015 – 2017

ACADEMIC SERVICE

- **Tutorial Contributor and Presenter**, “Understanding Art & Culture,” ACM International Conference on Multimedia Retrieval (ICMR), Amsterdam, Netherlands, 2026. Contributed and presented a segment on Chinese calligraphy as a case for multimodal art understanding, covering aesthetic structure, implicit micro-actions, and human-AI co-creation.
- **Web Chair** of The 16th International Symposium on Visual Information Communication and Interaction, VINCI 2023, Guangzhou, China
- **Serving as reviewer for journals and conferences** in visual computing/graphics & imaging, information visualization and HCI, AI and society, and electronic & information engineering:
 - CONF.: European Conference on Computer Vision (ECCV); ACM International Conference on Multimedia (ACM MM); International Symposium on Visual Information Communication and Interaction (VINCI);
 - JOURN.: International Journal of Human-Computer Interaction (IJHCI); The Visual Computer; AI & Society; Frontiers of Information Technology & Electronic Engineering; Visual Computing for Industry, Biomedicine, and Art (VCIBA).

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant / Guest Instructor

HKUST(GZ)

- *Introduction to Experimental Animation* 2023
 - Assisted in course design, delivered lectures;
 - Guided students in stop-motion animation shooting, simple animation producing, video editing and creative coding in visual computing projects.
- *Interactive Storytelling* 2024
 - Supervised student projects combining narrative design, human-computer interaction, and generative AI techniques;
 - Provided hands-on training in relevant software tools and digital media techniques, improving students’ production skills.

Research Mentor in MPhil’s Program,

HKUST(GZ)

- *Research and Artistic Creation on the Aesthetics and Dissemination of Chinese Calligraphy and Scripts* 2023–2025.
 - Supervised postgraduate research on computational interpretation and aesthetic modeling of Chinese calligraphy;
 - Guided interdisciplinary collaboration between visual computing and digital humanities.
- *Multi-Modal Embodied Human-Pet Interaction via Pet Wearable Devices* 2025–present
 - Managing an interdisciplinary research project integrating embodied interaction design, multi-modal data analysis, and AI-driven affective computing for human-pet communication.

INDUSTRY R&D EXPERIENCE

Network R&D Center, China Telecom Research Institute Beijing/Guangzhou, China
Applied R&D Engineer 2021 – 2022

- Built an LSTM-based traffic-flow forecasting model for short-term prediction from real-time sensor data.
- Developed a reinforcement-learning model for signal-timing optimization to improve traffic-light control at intersections.
- Trained license-plate detection and recognition models on in-house data, using YOLO for

detection and an OCR model for plate reading.

- Helped develop the MQTT data pipeline, including publish/subscribe workflows and connection handling, to feed models with real-time edge and roadside data; also contributed to front-end product design and requirements breakdown with product and platform teams.

Cloud Network Institute, China Telecom Research Institute Guangzhou, China
Cloud Network Researcher 2020 – 2021

- Drafted an initial ITU-T testing and monitoring framework for cloud-network architecture, translating requirements into test methods, test cases, metrics, and a standard reporting format through internal review and revision.
- Built a Prometheus-based prototype to run the framework in practice, supporting batch test execution and longitudinal key-metric collection.

GF Securities Guangzhou, China
Quantitative Researcher 2018

- Researched low-risk asset allocation for fund management using regression analysis and linear optimization, translating risk and compliance limits into a workable strategy.
- Built a prototype backtesting and risk-control system covering data processing, machine-learning factor extraction including LSTM, and strategy execution against historical data.

PUBLIC ENGAGEMENT, OUTREACH & LEADERSHIP

STEM outreach Lecturer — *3-Day AI Workshop for Teenagers* 2026
 Guangzhou, China

- Topic: how AI understand world — hands-on demos linking machine perception with creative practice;
- Designed and delivered a 3-day lecture series introducing AI concepts to high school students.

Invited Speaker — Salon: “*The Aesthetics of Innovation and Entrepreneurship in Nansha*”
 Guangzhou, China 2025

- Topic: “Seeing–Feeling–Gaining” perspectives on Nansha’s innovation and entrepreneurship ecosystem.

Invited Speaker — “‘*Qing Chu Yu Nan*’ Talent Gathering Action Plan” Launch Event
 Guangzhou, China 2025

- Topic: Youth-friendly innovation ecosystem in Nansha; tech integration (e.g., policy support for researchers).

STEM Outreach Lecturer — Tongren No. 1 High School 2023
 Tongren, Guizhou, China

- Designed and delivered two short courses: (1) Artificial Intelligence Technology and Calligraphy Appreciation; (2) Sustainable Green Building and Future Urban Development.
- Led hands-on activities (e.g., AI demo for calligraphy style recognition) and facilitated a student showcase.

Founder — Campus Fitness Community, HKUST (Guangzhou) 2022–Present
 Guangzhou, China

- Founded and led a 300+ member student-staff community; organized group workouts and wellness events.
- Coordinated scheduling and safety protocols with the university gym; supported facility operations.
- Promoted campus wellness initiatives and peer coaching.

Solution Presenter — ODCC Edge Computing Conference 2021

Beijing, China

2021

- Talk: “China Telecom Cloud–Network Convergence Innovation Platform: R&D and Application Practice.”

Director — Student Art Union, South China University of Technology

Guangzhou, China

2017–2018

- Organized and executed campus performances and events;
- Coordinated cross-department collaborations;
- Supervised teams’ daily training and budgeting.

LANGUAGES

Mandarin, English, Cantonese, Japanese (N2 Level)

HOBBIES

Bamboo Flute, Calligraphy, Trekking, Triathlon, Museum, Urban Exploration